Santhosh Vodnala

(+91) 9741607475 | santhoshv.iiitb@gmail.com | Santhosh Vodnala | vodnalasanthosh47

EDUCATION

International Institute of Information Technology Bangalore **July 2023 – July 2028**
BTech + MTech (Dual Degree) in Computer Science *CGPA: 3.66/4.00*

PROFESSIONAL EXPERIENCE

Summer Research Intern **June 2026 – Present**
COMET Lab, IIIT Bangalore

- Emulated a **9-host, 6-switch** topology in **Mininet**, configuring **6 realistic link profiles** (datacenter, backbone, home fiber, LTE, lossy Wi-Fi, satellite) with distinct bandwidth, delay, loss, and queue size configurations.
- Migrated video streaming from raw UDP to **SRT** (NAK retransmissions, **1s latency buffer**) to eliminate frame corruption on lossy links; an adaptive-bitrate prototype lifted mean VMAF from **33.5 to 50.0** on the worst link.
- Collected per-second QoS metrics (kernel **rx_bytes** throughput, ping-based latency, jitter, and loss) for **28 video streams** and aligned with frame-level scores from **Netflix's VMAF** model to build a **500+ row** QoS-QoE dataset.

UI/UX Design Intern **May 2024 – July 2024**
Think Logic Tech Solutions Private Limited

- Architected the **end-to-end UI and UX flow** in **Figma** for a comprehensive PG management platform, designing **12+ screens** spanning the hero page, user onboarding, and core functional toolkit.
- Collaborated directly with the founder to translate high-level business requirements into an intuitive design, reducing ambiguity in feature scope by aligning on **3 key user personas**.
- Partnered with the engineering team to ensure a seamless handoff, **delivering developer-ready component specifications** that reduced communication back-and-forth by an estimated **40%.**

PROJECTS

AmazSearch | Google ADK ***June 2026 – Present***

- Architected a **5-stage multi-agent** orchestration pipeline using the **Google ADK** and Gemini, transforming broad natural language user requests into spec-validated Amazon recommendations, **automating product research**.
- Engineered **specialized agents** for product search and compliance verification, leveraging Pydantic schemas for strict cross-agent data-flow validation.
- Developed automated script using SerpAPI to search and analyze **20+ Amazon listings per product query**, parsing product page details and top reviews to surface key customer pain points.

CollabGPT | Next.js, Google Gen AI SDK ***October 2025 – November 2025***

- Engineered a full-stack collaborative AI platform using **Next.js** and the **Google Gen AI SDK**, addressing core UX limitations of traditional chatbots such as context loss and poor shareability.
- **Led frontend architecture decisions** for a 3-member team regarding the **interactive node-based UI** with React Flow, enabling users to visually construct and manage complex multi-model conversational context.
- Implemented **state management** and **API integration**, preserving context across **4 node types**.

Low-level MIPS Processor Simulator | Python, OOP ***December 2024 – January 2025***

- Modeled a **single-cycle MIPS datapath**, replicating standard control-unit signals (`reg_write`, `mem_write`, `branch`, `jump`, `zero`) and hardware-accurate register-file ports across a **32-register** file.
- Decoded **25+ R-type, I-type, and J-type instructions** — arithmetic, logical, shift, branch, and byte/ halfword/ word load-store — directly from 32-bit binary encodings via bit-level field extraction.
- Built **two's-complement arithmetic, sign extension, and 64-bit hi/lo register handling** from scratch to correctly execute compiled MIPS binaries for a power-of-two checker, factorial, and exponentiation programs.

TECHNICAL SKILLS

Languages: C++, Python, Java, C, SQL, JavaScript

Tools, Frameworks and Libraries: Git, GitHub, Figma, Node.js, Express.js, React, Next.js, Tailwind CSS, EJS

Concepts: Data Structures, Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems, Database Systems, Programming Languages, UI/UX Design, REST APIs

ACHIEVEMENTS

- **Winner, AWS ImpactX Challenge (IIT Bombay Techfest):** Secured **1st place among 1000+ participants**, collaborating in a 3-member team to build the production-grade application CollabGPT.
- **Dean's Merit List:** Recognized for outstanding academic performance by IIIT Bangalore for Academic Years 2023–2024 and 2024–2025.